It’s Bugs’ World and We’re Just Living In It with Jessica Ware

Podcast: Startalk Podcast

AI Executive Summary

Core Thesis

The episode explores the ancient evolutionary success of insects and their critical ecological role as Earth's most abundant animals, contrasting this dominance with the severe threat they face from human-induced environmental changes such as urbanization, light pollution, and climate change. Through discussions on specific biological traits—from dragonfly eyes to lice evolution—and modern pest management issues like invasive species and mosquito control, the episode argues that insects are essential to biodiversity and that their declining population rates signal a crisis in habitat quality necessitating more nuanced conservation strategies than currently practiced.

Key Arguments & Data

- **Taxonomy Clarification:** True bugs (Order Hemiptera) possess sucking mouth parts (straws/pumps) and undergo incomplete metamorphosis; others (beetles, flies, grasshoppers) utilize chewing mouths or lapping tongues and exhibit complete metamorphosis.
- **Evolutionary Timeline:** Flying insects predate dinosaurs by approximately 100 million years, with the first wings appearing roughly 400 million years ago. The sky was originally empty of predators, allowing insects to evolve large size despite limited aerodynamic experience initially.
- **Dragonfly Biology (Odinatology):** Dragonflies have incredibly high visual processing needs; their brains are tiny relative to body mass but rely on massive optic globes for a mosaic vision. They possess internal air sacs (iron lungs) that allow for buoyancy and flight regardless of atmospheric oxygen levels, contrary to the historical theory that high ancient atmospheric oxygen was solely responsible for their size.
- **Mating Behavior:** Male dragonflies step on females during mating to ensure paternity; this often crushes the female's rear vision omatidia.

This evolutionary trade-off prioritizes reproductive success over sensory integrity.

- **Insect Abundance & Diversity:** Insects outnumber all other animal groups combined by both species count and biomass. Manhattan has its own unique species of pavement ants and New York City's insect biodiversity is largely extirpated due to human development.
- **Human Impact on Habitat:** Light pollution disrupts firefly communication and mate finding (e.g., "no fireflies in Times Square"). Noise pollution affects cicada emergence success. Urbanization creates artificial habitats for invasive pests like roaches and lantern flies while destroying native biodiversity.
- **Insect Decline:** Global insect populations are declining at unprecedented rates following the Industrial Revolution, with specific species (Monarchs) dropping 90% since 1970. Habitat loss, climate change-induced range expansion (ticks), and pollution are primary drivers.
- **Mosquito & Pest Control:** Bug zappers are inefficient for mosquito control (killing ~90% beneficial insects instead of the target species). Sterile male release programs exist but struggle with scale due to mosquito breeding ecology compared to dragonfly nymphs.
- **Lice Evolution:** Head lice and genital/body lice are genetically distinct, speciating around when humans became hairless or developed specific hair patterns (30,000 years ago), rather than spreading from one species to another via human contact myths. Lice infestations correlate with hair texture rather than ethnicity (lice thrive in straighter hair).
- **Insect Diet & Edibility:** Insects are generally high protein/low fat, but some varieties (e.g., cockroaches/water bugs) are very fatty and not suitable for consumption compared to grasshoppers or crickets. Cockroach populations include many native species plus those brought over the Transatlantic slave trade, with heat-induced flying behavior.
- **Haldane Quote:** J.B.S. Haldane stated that God must have an "inordinate fondness for bees/beetles" due to their abundance and evolutionary success.

Actionable Takeaways

- **Mitigate Light Pollution:** To protect fireflies and other night-active species, residents should reduce unnecessary lighting in outdoor spaces (yards, lawns) to restore natural mating signals.
- **Avoid Ineffective Pest Control:** Discontinue use of bug zappers, as they kill far more beneficial insects than mosquitoes; consider manual removal for specific pests or targeted control.
- **Support Native Flora:** Protect milkweed habitats to help Monarch butterflies and other pollinators survive their long migratory routes (e.g., planting native species in yards).
- **Early Detection of Invasive Species:** Learn the local insect inventory; early reporting of unfamiliar bugs (e.g., lantern flies, bull weevils) to municipal entomologists can prevent established invasive populations.
- **Address Climate Change Awareness:** Recognize that expanding tick ranges and habitat changes are linked to human-induced climate shifts; supporting general environmental health is essential for maintaining insect populations.
- **Educate on Myths:** Teach children that lice spread via head contact rather than hygiene status ("share the hat") to reduce stigma associated with infestations, particularly regarding hair type myths regarding ethnicities.
- **Protect Biodiversity First:** Acknowledge that insects are not just pests; protecting them is vital for ecosystem stability, food webs (pollination), and human health.

Full Transcript

Nagin, I was, like, weirdly attracted to these bugs that we were talking about. And I'm, like, emotionally connected to them. Ooh, how did that happen? Coming up on Star Talk. Welcome to Star Talk. Your place in the universe where science and pop culture collide. Star Talk begins right now. This is Star Talk. Neil deGrasse Tyson, you're a personal astrophysicist, and I got with me, Nageen Fursad. Nageen, welcome back. Oh my god, hello. What having you? I love being here on this hot summer day. Yeah, excellent. You know what we're going to talk about today? Tell me. Bugs. Oh, so delightful. Literally love it. When a bug lands on me, I just think, oh, this is great. Love a bug. Well.

I, not be an expert, but I work in an institution that contains them. Not only experts, but also bugs that they study. Back for your third time on StarTalk. Jessica, where? Jessica, welcome back. Thank you for having me. Can't get enough. I'm excited to talk to you. And one of those times we talked about cicadas. Yeah. because you're like one of the world's experts on these ugly bugs. I think it's pronounced beautiful and charismatic. I mispronounced it. Did you say they were charismatic? Yeah. Colorful, check. Sounds beautiful, check. They do cool things with the genitals, check. That's charisma. I mean, do they give enlightening stump speeches on the road in national politics? On tree stumps, yes. So let me get your whole title here. Division chair in the division of invertebrate zoology. So these are spineless animals. Exactly. Well, the people in my division or the...

No, no, no, no

metamorphosis. There's one group that does it hola metabola. I don't know that's the official term for that. Yeah, that's called hola metabola. It's a whole Metabola like it's a complete metabolic change of its body. I don't study those I study all the ones that don't do that. Oh, you don't? Oh, she just described what she doesn't study. Yeah. Yeah. Non holo metabola. So like a baby grasshopper hatches from an egg and just is a small version and it just molts to a larger and larger size until it's the same size as an adult grasshopper. There's no complete rearrangement of its body. You're also a PI, principal investigator.

The Institute for Comparative Genomics? Comparative. This is one of the jewels in the crown of this institution. It's true. We are able to really understand 400 million years of evolution through the genome, through whole genome sequencing. Cool. And your professor in our built-in Gilder Graduate School. Yeah. Loving it. Loving it. Yeah. So you're the one to talk about bugs. Well, insects. Okay. Bugs are just one kind of insect. Not to... Tell me that to all bugs. No. Not to be that guy. Stop! Stop! but also non-holding monadic or whatever, just so we're clear. Does she study non-bugs? Okay, let's just figure this out. What's a bug? So bugs are in the one particular kind of insect, and they're in the order Hemiptera. They're called true bugs. They're the only ones that are actually bugs. All the rest are just different kinds of insects. Dragonflies are dragonflies, beetles are beetles.

Bugs are bugs. They're just a kind of insect and they all are united because they have a mouth part that's sucking. They have like a little straw. It's so beautiful. A little straw and a little pump to suck up things. They suck up plant juice mostly except for the ones that suck up blood. Yeah. So are bees bugs? They suck up plant nectar? Bees don't. Bees have a tongue. Bees lap up plant nectar. They're like little cats. I didn't realize that. Their tongue is really long. It looks like if you look at a bee when it's drinking, it actually has a pretty long tongue and they stick it out and actually they're just, they're lapping up the nectar. Wait, wait, stop. You have divided yourselves among people who study insects who suck and insects who lick. Who lick, yeah. That's a category. So you don't study insects who lick? Yeah, no, none of the, I study insects that bite, that chew, and that suck. Wow, you just made a weird exception for that one category of licking. They're dead to me. Okay, I will not. No. Just like the holo metabola. She's a non homo

metabola. She's a non homo metabola. Yeah, non homo metabola. All right, so those are bugs. So how about when I'm outside in a twilight and gnats come around my head? Yeah. They're bugging me.

So to me, they're bugs. Are gnats bugs? No, they're insects. They're in the order diptera, which are the flies. So gnats are a kind of fly. There's gnats. There's noceums. There's horse flies. There's mosquitoes. We all know mosquitoes, black flies. And those are all in the order diptera. Wait, I heard noceum, and I didn't believe that was the actual name of an insect. I thought the person was just lying. Noceum. Yeah, and it's all one word. Noseum. Noseum. Do you know what? You don't see them. You don't see them. No, you don't see them. Finally, you named something after what it looks like. Yep. Because none of that other stuff, we don't know what you're talking about until you explain it. Yeah, the dragonflies, I mean, they don't really look like a dragon. They do fly. I'll give you that. Right? It's in the name. They do have like a dragon-vibe-ish look, though. Wait a minute. We eat them. Grasshoppers, they're often eating grass and they hop. Descriptive.

That's two that we know of that are very disruptive. Well, I bet you that your official name for a grasshopper has neither of those two in it. It's true. It's orthoptera. So there. What's wrong with you guys? I know. So who knew that bugs had an official definition? It's the first I've learned of this. I'm delighted to be enlightened this way. And I notice you have tattoos on your body. Don't tell me that they're what you study. I have a dragonfly, which is what I study, but I also have, well, I have a bug, a troop, oh, I have a troop bug. I have a hemiptera right here. Oh, troop bug, okay. With a sucking mouth part, because it's a sucking mouth part. And then I have a tiny rat flea. It's wearing a circus hat and riding on a penny farthing. Right here. And none of those licks. And none of those are licks. Gotcha. What accounts for, even though they're all different from each other? So I don't mean to say they're the same, but.

They're very different from us. What makes them so successful on Earth? Well, part of their success is the time. So they've been around for a really long time. Flying insects have been around for 400 million years. So they've had a lot of time to rapidly radiate into all the different niches on Earth. So they just predate dinosaurs. Yeah, predate by quite a bit. Oh, by 100 million

years, yeah. By quite a bit. Like, none of the cool dinosaurs that you might see on the fourth floor They weren't even on the scene yet when dragonflies were already flying. The first to fly, so they were able to kind of conquer that. Dragonflies are OG flyers? OG, they're the OGs. Yeah, before birds, before bats, before pterosaurs. So you realize in the sci-fi novel Dune, they've got, it's another planet, of course, and everything's different. And they're flying machines whose wings duplicate what a dragonfly wings do. So it's not a rotocopter.

and it's not an airfoil, but they look like dragonflies. So he must have been very impressed with them, just as you are. I mean, it's the original design, right? It's the bow plan, the template, right? Most insect wings follow the same idea. There's something slightly different that dragonflies and damselflies do with the way that their muscles attach to their wings inside of their body. But in general, the wings are the same. The forewing kind of has air that passes over it. It goes and it affects the wind that passes over the hindwing. And what were you doing just before you arrived at my office here? Oh, we were looking at dragonfly brains, brains. How big are their brains compared to their bodies? So you know, like when you look at a dragonfly, their head just looks like giant eyes. That's kind of the story. So their eyes take up almost everything, and then they have a teeny, teeny, tiny brain.

So they have a mushroom body, which we think is for memory. So they can remember some things and they have a tiny central cortex and then most of their head is just optic globes and because they're highly visual predators. So seeing is the biggest important part of what they do. And I mean, what can they remember? Like, can they remember? You know what I mean? Are they like, I can remember what I ate for lunch or are they like, I can remember like my family's story. Yeah, that's a good question. We think maybe what if you were territorial? and you had to remember where your territory was, would your brain be bigger than if you weren't territorial? Because some are territorial and some are not. Or some like never leave the pond from which they emerge, and some migrate across the ocean. So if you're migratory, would your brain be bigger than ones that never leave the area where they're born? So the eyes of these flying insects are completely different from human eyes, optically, right? And famously they're portrayed

as You know, in cartoons, I guess. That's where I get my entomology from. Where they're looking through and you see like a hundred versions of whatever it is they're looking at. They've seen it for house flies and dragon flies. They have the same kind of eyes. Yeah, they're basically the same idea, although the number of omatidia varies. There's an omatidia, which has a lens and it has photoreceptor cells. And there are lots of those little images, but we think that they get stitched together in the brain and kind of make one composite image that probably they're seeing. Is that the same for butterflies and moths?

We think it's pretty much the same across insects, but the difference is that some insects that have invested a lot in sight, like praying mantises, that's one, dragonflies or another, their vision is probably better than a butterfly or a moth. Insects also have eyes that sort of stick out. Does that mean they see top and bottom at the same time? Or what does that mean? No, that's true. That's such a good observation because the shape of the eye and the way that the omatidia, which are the lens plus the photoreceptor, the photosol. The way that they kind of project out, stick out, that actually affects whether they can see above them, a little bit below them, a little bit behind them. So dragonfly eyes are really globular and that allows them to see a little bit behind them in case there's a predator coming or there's food or a mate and they can see out laterally and they can see in front of them and a little bit above them. Except for like they do something with sex that's a little bit weird and males grab the female behind the back of her head.

And when he does so, sometimes he steps on her omatidia, so she loses a lot of her vision in the back part of her head. What? And that's built into their process? That's what they do, yeah. What is the evolutionary purpose of doing that? Well, if you hold on to her, then she will fly away from him and then he can ensure paternity. So getting pregnant has higher evolutionary value than losing some of your sight in the back, I think. Because not everything is It's good. It just has to be less bad than something else. If your fitness is even a tiny bit higher because you pass more genes on for the next generation selection. Yeah, it'll accumulate and then... So you can tell when a dragonfly is female because she's the one that's always like...

I'm Joel Cherico and I support StarTalk on Patreon. This is Star Talk with Neil deGrasse Tyson. And that, so they all share the same things. Of course, but somebody's gonna have the bug eyes first, and then does that give all the bug eyes? Or did eyes evolve separately to be similar? No, we think that there's a single origin. Convergent evolution, I think you guys call that. Yeah, but I think there's a single origin for eyes, a single origin for wings. And so insects, in general, have a head, a thorax, and an abdomen. They all have that. The abdomen, in general, has ten segments, although this... Oh, so they're like, they're doing a ten pack. They're doing a ten pack. The common ancestor all had a head, a thorax, an abdomen, and six legs. Got it. Yeah. The basal hexapods, which are the closest relatives to dragonflies and damselflies, they had no wings. They never evolved them.

And then at some point, 400 million years ago, boom, wings. Wow. Boom. I think that was, I believe that was the sound it made. I got wings. But you have to learn how to fly. That can't have been good. That must be like a real head trip when you're like, suddenly I have these things. What do they do? I think they were really, what we think is that they were not good flyers in the beginning. Cause the ones that first, like the sky was empty, right? You could be as big as you want. There was nothing else really hunting you except for each other. So the first ones that flew were really large. Each wing was like 37 centimeters.

They were pretty big and they were really clumsy. They had really stiff wings and they were just kind of like Probably not great, but then as as the sky filled up with other things then wings tended to get a lot smaller and there's a bunch of different parts of the wings for fast flight for maneuverable flight for aerodynamics and for hovering again for hovering because dragonflies hover Yeah, they I mean they're not as good as hover flies or for some other there are some it for the pesky entomologists is listening to this they'll say They're not as good at hovering as some other things, but they can stay in one spot for a little while. Right, right. And then there's something like crickets that can't fly.

Well, crickets can fly. They just more like glide. They do like a little hop and they kind of glide. Right. So is that just because like, is that like they started with that evolutionary thing, but then they just didn't go forward

with it? They were like, we don't really need that model. Yeah. I mean, they have the ingredients in the kitchen, right? They have the tools in the toolbox, but they're really focused on jumping. Dragonflies are going everywhere. They're going around the whole globe. so their wings are kind of designed for long distance flight. Wait a minute, you're saying they got big because no one was in the air to eat them, like a bird or something? Well, there's a couple reasons why we think they got big, but the reason why they were able to be big and clumsy was because there was nothing else in there related to... What's this I remember about... because there was a time when there was much more oxygen in our air, and insects breathe through their skin, right?

Well, they have little spiracles. They have little holes that open and close and let air go in. And we used to think it was just that high oxygen, so they were bigger. That's what I read, like long ago. Well, people have even reared dragonflies in high oxygen environments, and they did end up significantly bigger than if they hadn't reared them in the high oxygen environments. But recent work, just like off the presses this spring, not from our lab, but from collaborators in Odinatology, they showed that they actually have giant kind of air sacs in their body. They actually held air with them. And it really wasn't necessarily because the oxygen was really high that they were able to be big. It's that they actually were able to carry their own air with them. Oh, it's like a built-in oxygen tank. Yeah, they had their own iron lung. Oh, yeah, that's nice. What did you call that field? Odinotology. Odinata is. That's dragonflies and damselflies. Yes. And so if you study Odinata, You are an Odinatologist. Yes. And your field is Odinatology. Got it. Yeah. I had never heard of that before. So that's nice. If you Google it, you often get dentistry things. Because there's another word that's related to dentistry that sounds very similar. So just like etymology, which is words and entomology are similar. That gets confused if you do a quick Google. Yeah. And Odinatology and Odinatology, you'll get those sometimes you get hits for both. So be careful.

Yeah, yeah. Be careful when you're googling. I will totally be careful next time. You might end up thinking that insects have a lot of teeth. Or look, you want to see a beautiful picture of a dragonfly and then you get a picture of a roo canal. You don't want that. You don't want that. So insects are all

built on a basic model that apparently is highly successful. And they're more insects in the world than any other kind of animal. That's a true fact. In terms of abundance, that's true. Yeah. And in terms of number of species. Also true. Especially given how many species there are, that's the evidence of how successful. Yeah. I mean, there's around the same number of species of dragonflies and damselflies as there are all of mammals. So if I asked you to name five mammals. I don't like that. Well, I mean, it makes sense though, because if you live in, I live in New York City and I don't like ever run into like a fox or like, you know, a capybara, but I do on a daily basis run into insects. Yeah, you don't run into a capybara.

I don't even. Both mammals, both mammals, yes. You know, so the numbers kind of make sense because they can infiltrate like every world. So insects as a branch in the tree of life has the capacity to be highly fit in almost any environment it encounters. They're like gentrifiers. They move in, they take over. So I can move from the countryside into a city and they're insects in the city. Yeah, we have a lot of insects in New York City. Did any evolve here to be unique? We have a couple of species that we think are unique like Manhattan. Manhattan? Manhattan, which is a species that was described first here in New York City. Wait, what? Yeah, what is it?

It's good for all seasons. I just feel like when they're wearing a black Is that mostly in Central Park or is it? No, there's, I mean, you could just walk right down the street and there's pavement ants that are having battles on the street. There's pavement ants that actually fight each other. You can see dragonflies flying up Fifth Avenue. I mean, this city is actually full of little bits of green space, right? Whether it's a park, you know, like a little bits of flowers that you have on your stoop or a tree. So I used to host a program for PBS called Nova Science Now, the spin-off of Nova. And one of our episodes, we went to the High Line.

before it was heavily developed. The High Line had been abandoned for decades. It was just railroad tracks up there, okay? In Manhattan. And we go up there, there's like grass and there are insects. And I found a praying mantis. And I'm thinking, how does a praying mantis get into Manhattan? What did it does it hitch a ride on the on the ferry? Does it walk across the bridge? Manhattan is an island. It can do all of those things. It can also they

can they can glide with their wings. They can cling to a bit of bark and go across the Hudson River. They can hitch a ride on my car when I drive in from New Jersey. Like insects are highly mobile. And even though we think of.

praying mantises are not like long distance fires, they get around. Well, and do they not miss their families? You know what I mean? Are they like always like cutting and running and then just like going to Manhattan? I think about that all the time. If a fly gets into my car and then I don't get it out for another hundred miles, then it gets out. That's sad. Yeah, you're like you live in Pennsylvania now, buddy. Not a lot of insects get to see Pennsylvania, is that what you say to them? I say that sometimes when I'm bringing insects back from the field and I'm bringing them back to this building, I say, you know, not everybody gets to see New York. This is like actually a gift that gave them. That's to make her feel better before she puts the pins through its thorax. And murders them. Yeah, not a lot of insects get to see New York, right? So there's a lot of taxa that have been in the Northeast and species that have been in the Northeast for a really long time.

And I mean, like millions and millions of years, some of them persist, some of them we have extirpated. There's a lot of things that used to be in Manhattan that are not there now because of us. Extripicated. Out. So when we build environments, like the ones that we live in, things that end up being like cities, things that do well there are things that used to live in cliff. on cliff faces that used to live in areas where there was lots of impervious or hard surfaces. Like the rock doves, which we call pigeons. Like rock doves, like pavement ants. Pavement ants didn't evolve. It's called pavement ants. Prehistoric pavement. But probably they were just on rocks. We basically made a rocky environment that a lot of species like to live in. And which ones have we extirpated? There's a lot actually. We probably don't even know all the ones that were here because it's not like when people were first settling.

In Manhattan and this is millennia ago. They were keeping meticulous logs of biodiversity per se and the one Manhattan was colonized You know, I don't think people were either taking meticulous logs. There was a first entomologist Thomas say in North America that he actually did keep

records of all the things that were around so because of that We know that there are some cicadas that used to be here that are not here. We know that there's certain species They couldn't take The cicadas ironically couldn't take the noise It was too much competition with what they were doing. They couldn't make it here. So they couldn't make it anywhere One of the more successful bugs in New York and I won't have the right word but cockroaches tell me about So they live with us. Yeah, they didn't they didn't create their own world They are sharing our world and bedbugs. Okay? Yeah. So is this adaptive, I guess? There's only a tiny fraction of cockroaches that live with humans, right? And most of them... Don't tell me that that's only a tiny fraction of all the cockroaches. Because it feels like all the cockroaches.

When you like walk down the street to Manhattan at night, you're like, they're literally all here. No, there's like over 7,000 species in the order Bladodia, which are cockroaches and the social cockroaches, which are termites. Those are all just one order. And most of them live outside of the human condition. They're living in forests and deserts and rainforest. So. There's just a small number that have adapted to live inside of dwellings or to live inside with garbage and refuse. It's not us. It's the sewage, right? Like, we make garbage, we make refuge. It's true. They actually don't want to bother you. Yeah, it's not us. They turn on the lights. They scurry. They scurry. And the big ones, like you guys have the big ones? The big, the big, what do we call them, water bugs? Well, they're not water bugs, because actually that's a classic landlord.

They'll tell you you have water bugs so that you don't freak out and make them spray. But water bugs is actually a bug. It's in him if it's a totally different insect. But the ones that are big... A water bug is a completely different insect from a cockroach? Yeah. So if your landlord tells you that you have water bugs... But it squashes under my foot in exactly the same way. Also, are all landlords entomologists? And I didn't know. I have had landlords. When I had cockroaches, the landlord told me those are water bugs. I was like, surely you just. I am entomologist. I know this is not a water bug. It's not in Emitra. Hello. It's in Blaudea. And he said, how do you know you didn't bring them from work? Oh. No. And I was like, I don't actually know. He makes an excellent point. Yeah. Because it's also

possible. Plus, if you go to her section of the museum, there's signs that say, No exterminators. We do. We do exterminate in some parts, but we actually have certain rooms that we don't exterminate because we have live intentionally live insects in there. The big cockroaches we have though, they are not native to here. The ones that are the tiny little guys. We grew up with really big ones in Palm Springs, California in the desert. And they would sometimes like take flight a little. Yeah, the heat if it's hot.

Even the ones that we have here, they'll fly. But they're actually originally from Africa. And they were brought over on the transatlantic slave trade in large numbers. So they're actually like, they're adapted to be in hot environments. And when it's hot, they'll fly. So some of those people will be really freaked out. This is like one more new fear. Flying roaches new fear unlocked traumatizing just just stop. Yeah, if you were like not like you should care about climate change We all should but if you were like I'm on the fence about Then it as it gets warmer cockroaches will take me guys. Let me go there. Okay, two things first I heard tell That the world population of insects is on the decline. Yeah, my first inquiry with you does that relate at all?

to whatever impact climate change has on the world population of insects, on bugs, and do they migrate? Do they hang out? Do they spring forth? Are these connected? So insects, we think, are declining at a rate that we've never seen before. People after the Industrial Revolution, there's stuff in the literature that said, hey, I'm not sure, but I think this might have a negative impact on insects. And fast forward to the early 2010s, people started doing studies that showed that populations of a particular butterfly had dropped by 70%. Populations of this other species dropped by big numbers. Bug zappers and raid spray. They're doing their job. Oh my gosh. Are they really? No. No. You know what it is? It's us.

It's we change the habitat, we increase the temperature, we have now extreme weather, there's flooding, there's increased fire. All of those things are things that insects didn't involve with, right? So we know that insects are declining at a rate that's faster than we've ever seen, but the problem is, is that all life that we think about, with the exception of maybe fish, right? It came after insects, right? So plants, they need insects to pollinate them.

Birds eat insects. fish eat insects you know mammals eat insects we eat insects in the absence of insects what did she just say I know I'm like we eat insects well there was a trend and maybe this is like leading to the kind I don't know of like insect snacks as like because they have a high source of protein yeah man what do you think of that trend I heard tell they're high protein Yeah, they are high protein, low fat, depending on what you're eating. The water bugs in your, the cockroaches are actually very fatty. So if you were going to skip one, I would skip those ones. But like... Okay, thanks. I'm writing that down. I'm writing it down. Cockroaches, not good for your diet. That's the one reason to not eat them, really. It's just about the calories. When you think about it, it's really about that. But I would say like, my finger is on the pulse of tomorrow. I'm a young hip urban...

woman and I think eating insects is cool and people should all do it. Well you're not concerned about us declining the population further by also eating them? I would not go eat street insects. I'm not gonna go eat insects. just like go and collect on Fifth Avenue. People, when they rear insects for protein, for cricket cookies or whatever, they're doing it in the same way that we rear cattle. Some arms, yeah. You just said cricket cookies, like it's next to the Oreos and the chocolate chip on the shelf. Yeah, you're selling it on the bake sale. Shouldn't it be? Cricket cookies. Okay. I confess I've had sauteed crickets. They're popular in Mexico, right? I don't know. It was served to me once. It tasted like chicken. They're like kind of nutty. No, they tasted like a snack. I mean, just snack like you'd have at the bar. What I don't understand is we're not much different other than the climate change part of this. We're not much different than we were 100 years ago, moving into cities, taking over the landscape.

Why is there extra loss of insects today? I think urbanization is increasing at a rate that's much different than it was 100 years ago. All that means is we are living on less and less of Earth's surface area. But we're expanding. Won't the insects like that? I think we're expanding, though, the amount of impervious surface, the amount of concrete, right? We're decreasing the amount of forest. We pollute the water. There's a lot of insects that are aquatic that dwell in fresh water. That kind of destroys their habitat. I was just recently driving, I forgot where. And I just noticed, I never thought to notice this before, I'm on a four lane freeway going in one direction, there's

a big divider, and then there's four lanes coming the other way, and I'm just looking at the total surface area that that is, it's huge. It's huge. And the lights that go over it, because we're gonna have lights so that we can drive at night, that negatively impacts insects. Why? Well, there's insects eat less, caterpillars eat less, moss that communicate.

They're flying, kind of using the UV light to navigate with, fireflies that communicate with each other at night. Light, bad, bad vision. But aren't bug zappers evidence that we're smarter than insects? Because we know they're sensitive to UV. So let's make a UV light and then zap them. But we're dumber than we think, right? We shouldn't pat ourselves in the back because we designed those to get rid of mosquitoes and biting insects and it actually like 90% of what bug zappers kill are beneficial insects. So they kill almost no mosquitoes. So you actually, there's no sense in buying it. So in a way, I've kind of felt like that's kind of a loss. Why can't we just get the bug zapper and then collect it all and just eat? Yeah, yeah, yeah. And the bug zapper also barbecues them. And then turn them to a delicious treat. Yeah, just a little channel goes over to the grill. Yeah, yeah. In a way that maybe it kind of also friccacies them itself a little bit with it. It does a little saute. A little sass barbecue. But also we're so bad at killing like the bugs that seem to be irritating everybody like ticks.

Oh, so yeah, so ticks are not bugs, they're arachnids. It's not that we're bad at killing ticks, it's that because of climate change, because of changing temperatures, ticks from other regions are moving and they're expanding their ranges. That's why we have the Lone Star tick here, for example, which is the one that causes a meat allergy. It can carry a pathogen that can cause a meat allergy. It's not, that's because of us. We changed the climate and now they've been able to move into that area. So we've created habitats that previously were hostile. to some varieties of insects. Now they can hang out and do the backstroke and now we're affected by that because that was a place we created. Yeah, we created them. The catastrophe that we're seeing, we created and we're kind of keeping it going. We could make decisions that would actually have really beneficial impacts on insects that we should make. Why don't we make it? I can't just go past this because I know people who've been bitten by the Lone Star tick.

They can't, so it's not that they can't eat meat, apparently, they can't eat mammals. So they can eat chicken and fish, so I heard. You know what I think? I think vegetarians created this in the lab. Yeah. They were like, what could we do? Alpha Gal, that's it. Yeah, but more like pescatarians created this in the lab, like a really specific subsect. So I hear tell that in Martha's Vineyard, which is, you know, island off of Cape Cod, you know, a destination for the rich. There's an airport there, you could take your private plane on to. As one does. Famously, don't the Obamas go there? As one would do. I'm told that tick was so prevalent on that island that stores are now reducing their supply of meat for sale because no one's buying them. And restaurants have menus for those who are thusly affected.

Specifically a cardiologist like are these maybe these ticks are out for our health. Yeah, they're a gift. Red meat consumption. I mean and nobody did it and they're like, what could we do? Cardiologist was like listen for some of the stuff from this drop in population. Is it just habitat or is there other things that affect them? Are they not finding mates? There are a lot of ways you can drop a population. So a population is just a single species. So we're looking across, it's true across all the species of insects that we've looked at, right? So what we think is that for some things it's water pollution. For some things, and that means that they can't, you know, they don't, they die. So they're not able to survive. For some things like fireflies, if it's light pollution, they can't actually see each other to find a mate. So they're not able to reproduce. Like it kind of varies depending on what the species is. That's why there are no fireflies in Times Square.

I always wondered that yeah, there should be I mean they they make these flashes and the flashes are species specific They need to be able to see each other so they can see the flashes to know which species to mate with I grew up in the desert We didn't have fireflies and so then as an adult when I was here in the summers I was like oh my god These are the coolest things I've ever seen and still in the summer I look at them with such awe they make light with their bum I mean it's incredible but light we call it You know I heard that Yeah, it's a summer tree. But yeah, I mean, we should all be kind of sad, right? The things that we think about when we think about summer, like fireflies or monarch butterflies, like those are declining at such alarming rates. Monarchs have decreased by like 90% since 1970. You

know, when I was a kid, I saw them all the time. Now I hardly see any monarch butterflies. Yeah, it's us. We destroyed the habitat. They need milkweed. It's a specific plant that they've evolved to eat.

And they have a arduous long migratory route that goes down to Mexico and back and they get hit by cars and and and what have you so I mean we're making we've made decisions in the past and we're making decisions even right now that negatively impact insects all right, so I got it This is the 900 pound gorilla in the room. Yeah, okay some insects are prettier than others, to the non entomologist. You mean like headlice, that's a gorgeous one. I'm assuming. Neil is body shaming in sex, by the way. I just want to say that for the record. He's like the, yeah, gnats aren't looks maxing enough. The mean girl part of this. Nobody doesn't like ladybugs. They're nice. It lands on you. It's not going to bite you or sting you or inject you with venom. Okay.

And so I can list insects that emotionally we're cool with, okay? More humans have died at the hands of mosquitoes than have ever died in all wars we have ever fought against each other. If we could render mosquitoes extinct from this world, none of us would complain. What would you do as an entomologist? If by that you mean that we would get rid of a specific kind of anopheles and a specific kind of Aedes, Aedes aegypti, those are the deadliest animals on the planet. Those are the ones that vector malaria, dengue, chickengania, yellow fever. If we got rid of those two or three species complex, then yeah, I think that that would actually have immense help to the human population. But there's- While maintaining others. Yeah, there's thousands of other mosquitoes. So the bats have plenty to eat. Yeah.

bats, dragonflies, there's lots of things that eat mosquitoes and there's thousands of species of mosquitoes and so they live outside of the human condition, they're not vectoring diseases that affect humans. Those guys get to survive. Do you feel like there's a way to do that, that like we can get rid of the bad ones and keep the good ones? They've been trying, so people have tried releasing sterile males that are just shooting deads that have no ability to impregnate a female. They've tried it a couple different ways. One with sterile males, they were irradiated and one where they actually have a bacteria called wabakia that actually makes male. It's like a dracar noir and

the ladies just don't want to have anything to do with it. Is that what it is essentially? You know what it is? The wabakia makes it so that the wabakia is transmitted by females. So the wabakia changes it so that males are basically, yeah, males are very low in very, very low numbers and only females survive. So females aren't fertilized. No eggs, no babies. They tried releasing dragonflies years ago.

But dragonflies do eat mosquitoes in the juvenile stage and in the adult stage, but they don't like to take... Juvenile is still in the water? Yeah. Juvenile mosquitoes and... They're sitting ducks. Yeah. Or sitting mosquitoes. They're sitting mosquitoes. Right? Because they're just in the water like that. Yeah, with the little saipan. Still water, not moving water. Yeah, still water. And dragonflies, as babies, as nymphs, they will eat mosquitoes. But they don't like to breed in large numbers on command. So they could never release them in large enough numbers. Because that would be the solution. Just put dragonflies everywhere. Do you know the film The Healthstrom Chronicle? Maybe my finger's not on the pulse of tomorrow. Am I not a young hip woman? It was an apocalyptic documentary from the 1970s describing the future where insects take over the world and they just eat all of our food supply. The Healthstrom Chronicle. Yeah. Check it out. It is a... I mean people have... I've never forgotten it. I saw it like once in the 1970s. This idea that there's quote-unquote injurious insects, the ones that are because they like the same things we like. A lot of insects eat the same things we eat, that they would come for our crops, that they would take over, that they're out to get us. They're all locusts. This is the apocalyptic scenario. Yeah, but in reality, I think there's a very small number of stored food pests or grain pests that were worried about locusts. She won't number species, but that doesn't mean they're not out there, ready to eat all of our corn. Yeah, I mean, if you don't...

put all of your corn in one giant pile that's open to insects to go in there and eat it, then you're solid, you're good. And how about the bull weevil? Oh, the bull weevil. I mean, the bull weevil was an invasive species. Invasive species are also a source of insect decline because invasive species often can out-compete native species. I think a bull weevil might be the only insect, whatever kind of bug it is. Yeah, beetle. Beetle, the only one that has a blue song named after it. Multiple. because there was people to different

versions. Oh, okay. There's a Lead Belly one that's very famous. Yes, that's the one I'm sticking up. There's a Bessie Smith one, like multiple people did, because you can, it's a blues. Bo-wee-boo. Yeah, you can pop it in. Yeah, because if you're picking cotton and there's a Bo-wee-boo eating all your cotton. Yeah. Where's our current like, pop hits about lantern flies? I know, right? That's what we really need, because those are, I feel like, our modern irritation. Yeah. What's a lantern fly? Well, it's the...

Shall I? I think you shall. Well, is it fly or bug? Lantern bug? Lantern? It's actually a bug. That's very astute as in a fellow entomologist. Thank you so much. I was very astute to call him. And you know, some things are arachnids and some things aren't. It's another thing I wanted to teach you all. No, but the lantern bug is like a thing that eats a bunch of... Everything, like trees and whatever. Yeah, it's got a sucking mouth part. It's got a sucking mouth part, not a licking mouth. Now I'm licking mouth, so it sucks the plant juice. And it's been in like, it's been in Manhattan, it's been in New York City for a few years now. And we've been told to stop them every time you see one. And they're sort of dumb, like it's easy to stop them. Yeah. They don't like run away real fast or anything. You just like easily murder. You gotta watch out though, because if you murder the ones easily, you're actually Promoting fast ones if you kill all the slow ones because they're easy to kill. Yeah, you just made a whole generation of fast ones, right? They're like adapting to be exactly so what you should do is only go after the fastest ones But then in the meantime the slow ones are still eating all the trees or whatever I mean, okay. Yeah, what are they doing sucking the plant the plant sap, right? Okay, but as we do when we make Maple syrup. Yeah in a lot of ways they're maple syrup tappers, right? The population we think is basically becoming established in the Northeast We tried scraping off their eggs because that was the other thing Well, maybe we get rid of the eggs and that really hasn't done anything So sadly we think that this is an invasive species that much like the bull evil is just kind of here I stomped one this morning, should I stop stomping them? You stomped the bull weevil? Where do you live? When I was out in the field. When I was riding my blues songs. We don't know that it's really making much of an impact. So the best, if you were to do anything, you know what you should do. As a New Yorker, as all New Yorkers should do, is we should learn the insects in our neighborhood so we can prevent the next.

invasive species from becoming established because if you knew what was like on your walk to work Right, and you saw something that what you weren't supposed to see that was unusual You would tell someone sooner and then you could prevent them from becoming established. So if the pavement is start lifting the slabs They're like jacked They work out in the yard All right, so a brief follow I think you already went there, but I'm gonna just to bring it closer to it. We have people among us who are saying, save the whales, save the porpoises, save this, save that. Nobody's ever said save the rats or save the leeches or save the, any other parasitic. Save the maggots. The maggots, there you go. If you will. Thank you. Save the head house. So what I'm asking is, we are very species biased about what it is we want to declare that we want to save, whereas all of these creatures are on God's green earth just trying to make a living. Yeah. It's not their fault that it's their job to suck your blood. Yeah. That's not, don't blame them for that. Yeah, the bed bugs just trying to make it in New York City. And they're making it. What did they do before mattresses, by the way, bed bugs? I mean, I feel like the day mattresses were invented, bed bugs were like, guys, this is my time. But like as if you look, it's actually, if you like, like reading about human history, people have been writing about bed bugs for a really long time. Because even when you had just clumps of straw, then there were bed bugs that were in there. And plus, lice, this is the most fascinating fact that I've learned about lice. Most fascinating, okay? Head lice are genetically different from genital lice. Yeah, it's true. Even...

Okay, no, no, and, and so, how's that happen? Wait, wait, wait, can I check, can I check? Okay, so, go really far back, 30,000 years. We're hairy all over our bodies. What, the lice wouldn't know who did just hair everywhere. Then when we started getting less and less hair, then we had a hair gap between our genital hair and the hair on our heads, so that the speciation would have occurred around when we started becoming hairless in the major parts of our torso. Orphaning one branch of the body lice from the other. I read that, is that still true? I think that's true, and I think we think that one is more closely related to the lice that gorillas have, and one is more closely related to the lice that chimpanzees have. Wow. Which is which? I forget.

whether it was. I don't wanna know. What are you gonna do with this information? I have a seven year old daughter. I wanna like say something fun when I have to like. About who did Glyce suffer? About Glyce. About thanks. About Glyce. But you know what? Glyce is, you could tell your daughter this. Everybody eventually gets Glyce. People feel a lot of shame about it, but like little kids, but they put their heads next to each other. And if you have lights, it's not that you're dirty or I felt guilty because my mom said if you shared hats, you're going to get headlights. And there was a headlights outbreak in my grade one class, you know? And I was convinced it was because Jennifer and I had shared a hat at recess. But guess what? Now that I'm a growing up entomologist, you could tell your daughter.

It's just kind of a fact of lice. I see what you did there. So you contracted lice? Head lice, yeah. For the record, for the record, that she was like 50 years ago. Not like yesterday. That was 50 years ago. And I felt a lot of, I felt very stressed about it, but it turns out actually that my mother, who's a high school teacher, there was also a head lice outbreak at her high school. So now I'm like, it wasn't even me. It wasn't even Jennifer's hat. Okay, I read and have reported on the fact that lice infestations are 10 times more frequent among white children than among black children. And just wondering, were you white back then when you got the... It's like reverse racism with the lice. So they like, they will go in any hair. It's easier to find them in hair that is straight. And people used to think that And this maybe was true for a period of time from when Madame C.J. Walker invented hair pressing. That would have kept it out. So if you were using lye in your hair, or if you were using chemical pressing, as we called it, then you would, to straighten your hair, then you would have been less likely to have lice. The lice doesn't stand a chance. But if you were having natural hair, which I had natural hair, then you really should have your head checked for lice. So this actually really happened.

My school, they would have the moms from the neighborhood come in and do a life check. But the moms said to the three black kids, my twin and I and this other black kid, we don't need to check your hair because black kids don't get life. But I had life. I had it and they didn't check it. And then I was scratching my head at home and my mom, and we're all scratching her head. And then my mom was like, oh. And then she said, oh, we all have

lice. And I said, oh, no, they said we don't get lice because black kids don't get lice. And my mom was really mad. And she went and she yelled at the principal because it actually is it's not whether you have black hair or white hair. It's whether you chemically if you did a chemical perm, you also like if you were also straight, if you said the the Laos like straight hair better.

So it's a relative, it's not that black people don't get it, it's that it's just less likely. It's not that they like straight hair better, it's just it's easier for us to find it in straight hair. So we often have overlooked, like in my case, they could have checked my hair and they didn't. I had a lot of lice. 50 years ago, just to underscore this was 50 years ago. Is that the day you became an entomologist? Oh my gosh, I really didn't like it. But nowadays, there's actually a lot of things that you can do if you have life. So God forbid your daughter has life. Call me. Okay. There's like ways you can feed it without chemical insecticide. What is the size of the smallest and the largest bug?

Oh, that's a good question. So there are these little tiny, tiny, tiny things called fairy flies that are just a couple of millimeters, and they're really, really tiny. That's probably the smallest one. And they fly? They can fly, but I mean, come on. They're very, very tiny, so they're not going, they're not going long. There's other than by wind. They can be blown by wind. And the biggest ones, in terms of length, there's some really, really long stick insects, phasmatodea, in terms of weight. Stick ones, look. Yeah, like how long? Those ones that look like sticks on, like... tree sticks. There's ones that are longer than about as long as my arm here. Oh, wow. But then there are really heavy... It looks like in Brazilian rainforest kind of...

or Vietnam or Asia, like there's lots of tropical. Yeah, tropical. Although you can get some pretty big ones in Texas if you go back there. There's some pretty big ones in Texas. But then in terms of like mass, I would say there are some beetles that are really, really pretty heavy. Like, what are we talking? Like an orange and apple or like a? Like a, like a navel orange. The weight of a navel orange. What? That's wild. And how big? Let's say, How about this? Let's say naval orange in weight and size. That grosses me out so hard. No, it's so good. Dude, cause if you step on that it would go pop. But you wouldn't. But you wouldn't, of course. I, of course, possibly wouldn't, depending on the situation. I'm so sorry. Cause you, to see that,

you'd actually have, assuming that you spend most of your time in New York City, you would have traveled somewhere to see it. And you wouldn't be like, oh, I'm here now. Squash. Yeah, yeah, yeah. It's not like in my Manhattan apartment. You don't go out of your way. Yeah, right, right.

And I remember reading National Geographic and they had an issue on beetles. And one of them was the Goliath beetle. Yeah. That was something like this big. Yeah. I mean, I would say like they're the big navel orange size. Okay. And beetles bugs? They're not. They're Coleoptera. They're holometabolous insects that have an egg. Holometabolous. They're non-holometabolous. Egg, larva, pupa. complete rearrangement and then adult. And there's more Beatles than there is almost anything on earth described right now, although I think that Bees, ants and wasps are catching up, because people are describing all of those right now. Please give us Darwin's quote about Beatles, and we will end on that. Or Haldane, could I give you Haldane's quote? That there's an inordinate? Oh, did JBS Haldane? Yeah, okay. How about if I give that one? Okay. Although Darwin did work on Beatles, that was the thing you were conversing, but Haldane said that God must have an inordinate fondness for Beatles.

That's the quote I was looking for, Haldane. He's got some good quotes. Yeah, a page-turner. Well, Jessica, my gosh. I had all this pent-up bug energy. Can we do this once a year? Yeah, let's do an insect time. Yes. Once a year. I mean, your entire thing fascinates me. And you're interested in it, your origin story, which I've decided is with the lice. It's a lousy story. And it was 50 years ago. And you've got dragonfly earrings. I have two different kinds, dragonfly earrings. And I have a beetle. ring as well. I went full insect I went. I mean you're repping insects hard today and they need a little love. It's fascinating. When the insects take over, she'll be their queen. We need someone advocating for them because everyone really hates them. So this is like a good... They need you on PR. We should all love the insects. We gotta save the insects if we want to save ourselves. Well again thank you Jessica. Thank you. A friend and colleague just across the way here.

at the American Museum of Natural History, right up here on the Upper West Side of New York City. We are here in my office at the Hayden Planetarium, and we're going to come back to you with more insect queries,

for sure. And McGean, your podcast. My podcast is Fake The Nation. Have I ever been on your podcast? Subscribe. You have. You've been on it a few times. Not lately. But not lately. We're gonna bring you back. The people need to hear your takes. You don't call, you don't write, you don't text. I know. We will bring you back and you'll have, now that you have this bug energy. Yeah. Fake The Nation. And we can catch you. Are you a regular on Wait, Wait, Don't Tell Me? That's right. Yeah, tune in to Fake The Nation. Wait, Wait, Don't Tell Me. We'll find you there. We love you.

All right, this has been the bug edition of Star Talk. Neil deGrasse Tyson, your personal astrophysicist. Keep looking up or down at the bugs.